THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 9: **Symmetry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. After a rotation of 200° a regular 9-gon coincides with itself. This rotation is 5 times the smallest rotation. The smallest rotation (degrees) is: 1

(a) 50 **(b)** 40 **(c)** 30 **(d)** 70

2. The number of letters among O, X, D, M, T with BOTH a vertical and a horizontal line of symmetry is: 1

(a) 4 **(b)** 3 **(c)** 2 **(d)** 1

3. The sum of the angles of rotation of a square and a regular octagon (in degrees) is: 1

(a) 145 **(b)** 125 **(c)** 135 **(d)** 165

4. A figure has rotational symmetry of order 5 and one of its lines of symmetry. The total number of lines of symmetry it must have is: 1

(a) 6 **(b)** 5 **(c)** 4 **(d)** 7

5. A regular polygon has 6 sides. The number of different angles (less than 360°, other than 0) through which it can be turned to coincide with itself is: 1

(a) 6 **(b)** 7 **(c)** 5 **(d)** 4

6. A regular polygon is rotated again and again through its smallest angle 30°. The number of turns needed to complete a full turn is: 1

(a) 11 **(b)** 14 **(c)** 13 **(d)** 12

7. A regular 8-gon has side 5 cm. Its perimeter (cm) is: 1

(a) 35 **(b)** 40 **(c)** 48 **(d)** 45

8. The image of a point 8 units from a mirror line is how far from the original point (units)? 1

(a) 8 **(b)** 24 **(c)** 16 **(d)** 17

9. **Assertion (A):** A parallelogram that is not a rectangle or rhombus has rotational symmetry of order 2. 1
Reason (R): Such a parallelogram has two lines of symmetry.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The letter H has no line of symmetry. 1
Reason (R): The letter H has rotational symmetry of order 2.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A figure fits onto itself after turns of 45° and 135° but not after any smaller turn than 45°. Find its order of symmetry and show that 135° is a multiple of 45°. 2

12. A figure has rotational symmetry of order 5. List all the angles (up to but not including 360°) through which it can be turned to coincide with itself. 2

13. A regular polygon has rotational symmetry with the smallest angle of 24° and each side 6 cm. Find the number of sides and its perimeter. 2

14. A figure fits onto itself after turns of 60° and 240° but not after any smaller turn than 60°. Find its order of symmetry and show that 240° is a multiple of 60°. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A designer wants a symmetrical logo with 8 identical petals. Find the angle between neighbouring petals, the number of lines of symmetry and the total of angles around the centre if 8 petals are placed. 3

16. A regular polygon has 8 sides (8 is even). Of its 8 lines of symmetry, how many pass through two opposite vertices and how many pass through the midpoints of two opposite sides? 3

17. A regular 9-gon is rotated about its centre by 200° anticlockwise and the figure looks unchanged. Verify this using the order of symmetry and find the number of rotations of the smallest angle that make 200°. 3

OR

A designer wants a symmetrical logo with 9 identical petals. Find the angle between neighbouring petals, the number of lines of symmetry and the total of angles around the centre if 9 petals are placed. 3

18. A regular polygon has 10 sides (10 is even). Of its 10 lines of symmetry, how many pass through two opposite vertices and how many pass through the midpoints of two opposite sides? 3

OR

A regular 10-gon is rotated about its centre by 144° anticlockwise and the figure looks unchanged. Verify this using the order of symmetry and find the number of rotations of the smallest angle that make 144°. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A shop board carries the letters A, H, S and T. The students study each letter for symmetry. 5

(i) Which of the letters have a vertical line of symmetry?

(ii) Which letters have rotational symmetry of order 2?

(iii) Which letter has no line of symmetry?

(iv) Which letter has both lines of symmetry and rotational symmetry of order more than 1? State its number of lines of symmetry and its order.

20. For a regular 10-gon of side 6 cm: 5

(a) find the number of lines of symmetry,

(b) the angle of rotation,

(c) the perimeter,

(d) the angle turned by a vertex when the polygon is rotated 3 times through the smallest angle.

OR

Two regular polygons have 6 and 9 sides. Find their smallest angles of rotation, the difference, and the least angle (in degrees) that is a turn for both (the LCM of the two angles). 5

END OF QUESTION PAPER — Total: 40 Marks